1.1.3 Some Mathematical Functions

Mathematica includes a very large collection of mathematical functions. Section 3.2 gives the complete list. Here are a few of the common ones.

`Sqrt[`x`]`	square root ($\sqrt{x}$)
`Exp[`x`]`	exponential (e^x)
`Log[`x`]`	natural logarithm ($\log_e x$)
`Log[`b`, `x`]`	logarithm to base b ($\log_b x$)
`Sin[`x`]`, `Cos[`x`]`, `Tan[`x`]`	trigonometric functions (with arguments in radians)
`ArcSin[`x`]`, `ArcCos[`x`]`, `ArcTan[`x`]`	inverse trigonometric functions
n`!`	factorial (product of integers $1, 2, \ldots, n$)
`Abs[`x`]`	absolute value
`Round[`x`]`	closest integer to x
`Mod[`n`, `m`]`	n modulo m (remainder on division of n by m)
`Random[ ]`	pseudorandom number between 0 and 1
`Max[`x`, `y`, ...]`, `Min[`x`, `y`, ...]`	maximum, minimum of x, y, ...
`FactorInteger[`n`]`	prime factors of n (see page 419)

Some common mathematical functions.

The arguments of all *Mathematica* functions are enclosed in *square brackets*.

The names of built-in *Mathematica* functions begin with *capital letters*.

Two important points about functions in *Mathematica*.

It is important to remember that all function arguments in *Mathematica* are enclosed in *square brackets*, not parentheses. Parentheses in *Mathematica* are used only to indicate the grouping of terms, and never to give function arguments.

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This gives $e^{2.4}$. Notice the capital letter for `Exp`, and the *square brackets* for the argument.

```
In[1]:= Exp[2.4]
Out[1]= 11.0232
```

Just as with arithmetic operations, *Mathematica* tries to give exact values for mathematical functions when you give it exact input.

This gives $\sqrt{16}$ as an exact integer.

```
In[2]:= Sqrt[16]
Out[2]= 4
```

This gives an approximate numerical result for $\sqrt{2}$.

```
In[3]:= Sqrt[2] //N
Out[3]= 1.41421
```

The presence of an explicit decimal point tells *Mathematica* to give an approximate numerical result.

```
In[4]:= Sqrt[2.]
Out[4]= 1.41421
```

Mathematica cannot work out an exact result for $\sqrt{2}$, so it leaves the original form. This kind of "symbolic" result is discussed in Section 1.4.

```
In[5]:= Sqrt[2]
Out[5]= Sqrt[2]
```

Here is the exact integer result for $30 \times 29 \times ... \times 1$. Computing factorials like this can give you very large numbers. You should be able to calculate at least up to `1000!` in a reasonable amount of time.

```
In[6]:= 30!
Out[6]= 265252859812191058636308480000000
```

This gives the approximate numerical value of the factorial.

```
In[7]:= 30! //N
                  32
Out[7]= 2.65253 10
```

`Pi`	$\pi \simeq 3.14159$
`E`	$e \simeq 2.71828$
`Degree`	$\pi/180$: degrees to radians conversion factor
`I`	$i = \sqrt{-1}$
`Infinity`	∞

Some common mathematical constants.

Notice that the names of these built-in constants all begin with capital letters.

This gives the numerical value of π^2.

```
In[8]:= Pi ^ 2 //N
Out[8]= 9.8696
```

This gives the exact result for $\sin(\pi/2)$. Notice that the arguments to trigonometric functions are always in radians.

```
In[9]:= Sin[Pi/2]
Out[9]= 1
```

This gives the numerical value of $\sin(20°)$. Multiplying by the constant `Degree` converts the argument to radians.

```
In[10]:= Sin[20 Degree] //N
Out[10]= 0.34202
```

`Log[`*x*`]` gives logarithms to base e.

```
In[11]:= Log[E ^ 5]
Out[11]= 5
```

You can get logarithms in any base b using `Log[`*b*`, `*x*`]`. As in standard mathematical notation, the b is optional.

```
In[12]:= Log[2, 256]
Out[12]= 8
```

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